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Memory system

Abstract

A memory system including a plurality of nonvolatile memories and a controller. The controller is connected to the plurality of nonvolatile memories via a plurality of channels. The nonvolatile memory stores a first parameter indicating a delay value. The controller acquires the first parameter from the nonvolatile memory. The controller delays start timing of data transfer to the nonvolatile memory via a channel among the plurality of channels by a delay value according to the first parameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-197919, filed Dec. 12, 2022, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a memory system.

BACKGROUND

(3) In a memory system in which a controller is connected to a plurality of nonvolatile memories via a plurality of channels, data transfer from the controller to each nonvolatile memory is performed via the channels. At this time, it is desired that data transfer to a plurality of nonvolatile memories is appropriately performed.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a configuration of a memory system according to a first embodiment.

- (2) FIG. 2 is a waveform diagram illustrating operations of a memory interface in the first embodiment.
- (3) FIG. 3 is a flowchart illustrating training operations of a memory system in the first embodiment.
- (4) FIG. 4 is a flowchart illustrating training operations of a memory system in Modified Example 1 of the first embodiment.
- (5) FIG. 5 is a flowchart illustrating training operations of a memory system in Modified Example 2 of the first embodiment.
- (6) FIG. 6 is a circuit diagram illustrating a configuration and operations of a memory interface in Modified Example 3 of the first embodiment.
- (7) FIG. 7 is a waveform diagram illustrating an example of the operations of the memory interface in Modified Example 3 of the first embodiment.
- (8) FIG. 8 is a flowchart illustrating training operations of a memory system in a second embodiment.
- (9) FIG. 9 is a flowchart illustrating training operations of the memory system in Modified Example 1 of the second embodiment.
- (10) FIG. 10 is a flowchart illustrating training operations of a memory system in a third embodiment.

DETAILED DESCRIPTION

- (11) Embodiments provide a memory system capable of appropriately transferring data to a plurality of nonvolatile memories.
- (12) In general, according to at least one embodiment, a memory system including a plurality of nonvolatile memories and a controller is provided. The controller is connected to the plurality of nonvolatile memories via a plurality of channels. The nonvolatile memory stores a first parameter indicating a delay value. The controller acquires the first parameter from the nonvolatile memory. The controller delays start timing of the data transfer to the nonvolatile memory via the channels among the plurality of channels by a delay value according to the first parameter.
- (13) In the following, the memory system according to embodiments will be described in detail with reference to the accompanying drawings. It is noted that the present disclosure is not limited by these embodiments.

First Embodiment

- (14) In a memory system according to at least one embodiment, a controller is connected to a plurality of nonvolatile memories via a plurality of channels, and data transfer is performed from the controller to each nonvolatile memory via the channels, but data transfer is devised to be performed appropriately.
- (15) As illustrated in FIG. 1, a memory system **1** is communicably connected to a host **100** via a host channel HCH and can function as a storage medium for the host **100**. FIG. 1 is a block diagram illustrating a configuration of the memory system **1**. The host **100** may be, for example, an information processing device such as a personal computer, or may be a mobile terminal such as a smart phone. The memory system **1** may be, for example, a storage device such as a solid state drive (SSD), a universal flash storage (UFS), or an embedded multi media card (eMMC), or may be a memory card such as an SD card. The host channel HCH may be, for example, a wired communication path conforming to an SATA standard, a PCIe standard, or the like.
- (16) The memory system **1** includes a controller **2**, a plurality of nonvolatile memories **3_1** to **3_N** (N is an any integer of 2 or more), and a buffer memory **4**.
- (17) The controller **2** can be connected to the host **100** via the host channel HCH, and can transmit and receive commands, data, and the like to and from the host **100**.
- (18) The controller **2** is connected to the plurality of nonvolatile memories **3_1** to **3_N** via a plurality of channels CH₁ to CH_N, and can transfer commands, data, and the like to and from a nonvolatile memory **3**. N is any integer of 2 or more. Each of the plurality of channels CH₁ to

CH_N may be, for example, a wired communication path conforming to the SATA standard, the PCIe standard, or the like.

(19) The controller 2 is connected to the buffer memory 4 via a buffer channel BFCH, and can temporarily store data, management information, and the like in the buffer memory 4. The buffer channel BFCH may be, for example, a wired communication path conforming to the SATA standard, the PCIe standard, or the like.

(20) Each of the plurality of nonvolatile memories 3_1 to 3_N can store the data, the management information, and the like in a nonvolatile manner. Among storage areas in the nonvolatile memory 3, an area for storing the data is sometimes called a storage area, and an area for storing the management information and the like is sometimes called a management information storage area. The management information includes delay value parameters, which will be described later.

(21) Each nonvolatile memory 3 may be, for example, a NAND flash memory, a resistance random access memory (ReRAM), a phase change RAM (PRAM), a magnetoresistive random access memory (MRAM), or a ferroelectric random access memory (FeRAM). The following description will focus on a case where each nonvolatile memory 3 is the NAND flash memory.

(22) Each nonvolatile memory 3 includes a plurality of memory chips 31. FIG. 1 illustrates a configuration in which each nonvolatile memory 3 includes the four memory chips 31. Inside each memory chip 31, a unit in which data write and data read can be accessed collectively is a physical page. A physical block is configured with the plurality of physical pages and is the smallest access unit in which data can be independently erased. Each physical block has a plurality of memory cells. Each memory cell can store a plurality of bits. In each physical block, when each memory cell of a memory cell group connected to the same word line can store n-bit information, the memory cell group is treated as n physical pages. n is an integer of 1 or more.

(23) The buffer memory 4 is, for example, a volatile memory such as a dynamic random access memory (DRAM) or a static random access memory (SRAM). The buffer memory 4 may be implemented by a register circuit or the like configured with a latch circuit. The buffer memory 4 can temporarily store data, management information, and the like. The buffer memory 4 can be used as a working area used by the controller 2. The buffer memory 4 may temporarily store data related to a write process or a read process, may temporarily store data for an internal process such as garbage collection (compaction), and may temporarily store the management information such as address conversion information. Although the buffer memory 4 is provided outside the controller 2 in this embodiment, the buffer memory 4 may be provided in a control circuit 253 or in the host 100. In addition, the buffer memory 4 is not necessarily a volatile memory, and may be the nonvolatile memory.

(24) The buffer memory 4 includes a parameter buffer 41 and a data buffer 42. Various parameters (for example, a delay value parameter 41a, a parallel channel number parameter 41b, a data bus width parameter 41c, a frequency parameter 41d, and the like, which will be described later) are temporarily stored in the parameter buffer 41 by the controller 2. The controller 2 determines various parameters by performing training, which will be described later, and after that, writes the various parameters in both the parameter buffer 41 and the nonvolatile memory 3. Until the power is turned off, the controller 2 can then read various parameters from the parameter buffer 41 and use various parameters. Various data (for example, write data, read data) are temporarily stored in the data buffer 42 by the controller 2.

(25) The controller 2 comprehensively controls each component of the memory system 1. The controller 2 may be implemented as a controller package including, for example, a system-on-a-chip (SoC).

(26) The controller 2 includes a host interface (host I/F) 21, a processor 22, a buffer interface (buffer I/F) 23, an error correction code (ECC) circuit 24, a memory interface (memory I/F) 25, and a bus 26.

(27) The host interface 21, the processor 22, the buffer interface 23, the ECC circuit 24, and the

memory interface **25** are communicatively connected to each other via the bus **26**.

(28) The processor **22** comprehensively controls each component of the controller **2**. The processor **22** can be implemented as a central processing unit (CPU) or the like. The processor **22** controls the write process for writing data to the nonvolatile memory **3** according to a write command received by the host interface **21**. The processor **22** controls the read process for reading data from the nonvolatile memory **3** according to a read command received by the host interface **21**.

(29) The buffer interface **23** controls the access to the buffer memory **4** under the control of the processor **22**. The buffer interface **23** can transfer the data related to the write process or the read process, the data for the internal process such as garbage collection, the management information such as address conversion information, and the like to and from the buffer memory **4**.

(30) The ECC circuit **24** encodes and decodes data for error detection and error correction in the read data. The ECC circuit **24** encodes the data (the write data) to be written to the nonvolatile memory **3**. In addition, the ECC circuit **24** decodes the data (the read data) read from the nonvolatile memory **3**. The nonvolatile memory **3** performs the error detection and the error correction on the read data by the decoding. The ECC circuit **24** notifies the processor **22** of the error correction failure when the error correction fails.

(31) Any algorithm using a reed-solomon (RS) code, a bose-chaudhuri-hocquenghem (BCH) code, a low-density parity check (LDPC) code, or the like can be applied to an encoding and decoding algorithm by the ECC circuit **24**.

(32) The memory interface **25** performs interface operations with respect to the nonvolatile memory **3** under the control of the processor **22**. The memory interface **25** can transfer the data related to the write process or the read process, the data of the internal process such as garbage collection, the management information such as address conversion information, and the like to and from the nonvolatile memory **3**.

(33) The memory interface **25** includes a plurality of channel interfaces **251_1** to **251_N**, a regulator **252**, a control circuit **253**, and a transfer circuit **254**.

(34) The regulator **252** is connected between the power supply circuit (not illustrated) in the host **100** or the memory system **1** and the plurality of channel interfaces **251_1** to **251_N**. The regulator **252** is connected in parallel to the plurality of channel interfaces **251_1** to **251_N** via a power supply line **255**. The regulator **252** receives power from the power supply circuit (not illustrated) in the host **100** or the memory system **1**. The regulator **252** generates an internal power supply voltage of the memory system **1** according to the power. The regulator **252** supplies a power supply current corresponding to the internal power supply voltage to the plurality of channel interfaces **251_1** to **251_N** while controlling the internal power supply voltage of the memory system **1** so as to be equal to a target value. The target value is a voltage value necessary for an electronic component to which the power supply current is supplied by the regulator **252** to operate.

(35) The regulator **252** is set with a maximum value which can cope with a speed of current change. The speed of current change is hereinafter referred to as a current change rate. The current change rate can be obtained by dividing an amount of change in a current value by time. The current change rate can be positive or negative. A maximum absolute value of the current change rate which the regulator **252** can cope with is set as a limit threshold value. The higher the target value, the larger the current change rate of the regulator **252**. When the regulator **252** supplies the power supply current, if the current change exceeding the limit threshold value (the current change rate of the regulator **252** exceeding the limit threshold value) occurs, there is a concern that the regulator **252** may stop supplying the power supply current.

(36) The regulator **252** is also set with the maximum value of current which the regulator **252** can cope with. For this reason, in the following embodiments, the current change rate may be read as the maximum current value.

(37) The plurality of channel interfaces **251_1** to **251_N** are located between the regulator **252**, the control circuit **253**, the transfer circuit **254**, and the plurality of channels **CH_1** to **CH_N**. The

plurality of channel interfaces **251_1** to **251_N** correspond to the plurality of channels **CH_1** to **CH_N**. Each of the channel interfaces **251_1** to **251_N** is connected to the corresponding channel **CH**. Each of the channel interfaces **251_1** to **251_N** uses the power supplied from the regulator **252** under the control of the control circuit **253** to perform interface operations on the corresponding channel **CH**. For example, each of the channel interfaces **251_1** to **251_N** can transfer the data from the transfer circuit **254** to the memory chip **31** via the channel **CH**, and receive the data from the memory chip **31** via the channel **CH**.

(38) Each channel interface **251** may have a buffer memory **2513**. The buffer memory **2513** can temporarily store data transmitted to the channel interface **251**.

(39) When the power supply current is supplied from the regulator **252** via the power supply line **255**, the channel interface **251** can transfer the data to the nonvolatile memory **3** via the channel **CH**. The channel interface **251** can transfer data to the channel **CH** while temporarily storing at least a portion of the data in the buffer memory **2513**.

(40) The transfer circuit **254** is a circuit that receives write commands and write data from the host and transfers write commands and write data to each channel interface **251**. The transfer circuit **254** may be a component such as a shift register in which a plurality of flip-flops are connected in series in a plurality of stages. The transfer circuit **254** may be supplied with a clock from a timing generator (not illustrated). When the data includes a plurality of bits and the plurality of bits are stored in a plurality of flip-flops, the transfer circuit **254** may sequentially shift the plurality of bits between the plurality of flip-flops in clock synchronization. The transfer circuit **254** may sequentially output the plurality of bits bit by bit from a final stage flip-flop in the shift register.

(41) The control circuit **253** controls the channel interface **251**. More specifically, the control circuit **253** is a circuit that controls the start timing of the data transfer when the data transmitted to the channel interface **251** is transferred to the nonvolatile memory **3**. For example, when the access requests involving the data transfer to the two or more nonvolatile memories **3** via the two or more channels **CH** are generated in parallel, the control circuit **253** delays the timing at which the channel interface **251** starts the data transfer to the nonvolatile memory **3** via the channel **CH** by using the delay value according to the delay value parameter **41a**.

(42) The delay value parameter **41a** is data in which the channel **CH** and the delay value are associated. The delay value parameter **41a** includes the plurality of delay values. The delay value is a value for delaying the time from when the channel interface **251** receives the data from the transfer circuit **254** to when the channel interface **251** starts the data transfer via the channel **CH**. The delay value is associated with a combination of the channels **CH** on which the data transfer is performed in parallel. For example, when the data transfers to the channels **CH_1** and **CH_2** are started in parallel, a delay value **D1** is set. When the data transfers to the channels **CH_2** and **CH_3** are started in parallel, a delay value **D2** is set.

(43) The delay value corresponding to the delay value parameter **41a** is determined according to the speed of the control operation of the regulator **252**. The delay value according to the delay value parameter **41a** may be determined so that the current change rate from the regulator **252** is the limit threshold value or less. The limit threshold value is determined according to the speed of control operation that the regulator **252** can follow. Details will be described later.

(44) The delay value parameter **41a** is stored in the management information storage area of the nonvolatile memory **3**. When the memory system **1** is powered on, the delay value parameter **41a** may be read from the nonvolatile memory **3** by the controller and may be stored in the parameter buffer **41** of the buffer memory **4**.

(45) The control circuit **253** delays the start of the data transfer via a certain channel **CH** by a delay time converted from the delay value, and performs the data transfer via the channel **CH**. The conversion from the delay value to the delay time can be implemented, for example, by counting the number of clocks with a counter.

(46) Whether the data transfer between the channels **CH** starts in parallel (that is, starts the data

transfer simultaneously) may be determined by the difference in the start timing of the transfer of the control signal of the target channel CH, which is smaller than the response time of the regulator 252. When it is determined to be parallel (for example, simultaneously), the start timing of the transfer of the channel CH of which the transfer start is relatively slow or the designated channel CH may be delayed to the time longer than the response time of the regulator, or the time determined by setting.

(47) Next, operations of the memory interface 25 will be described with reference to FIG. 2. FIG. 2 is a waveform diagram illustrating current movement associated with the operation of the memory interface 25.

(48) The graph in FIG. 2 represents changes in magnitude of current flowing through the channel CH_1, the channel CH_2, and the power supply line 255 in order from the top. The current change in the power supply line 255 is the same as a current change in an output current of the regulator 252 that supplies power to the power supply line 255. A case will be described where the memory interface 25 simultaneously receives a write command WC1 for the nonvolatile memory 3_1 and a write command WC2 for the nonvolatile memory 3_2.

(49) At timing t1, the memory interface 25 almost simultaneously receives the write command WC1 for the nonvolatile memory 3_1 and the write command WC2 for the nonvolatile memory 3_2. At this time, the transfer circuit 254 almost simultaneously transfers the write commands WC1 and WC2 to the channel interfaces 251_1 and 251_2.

(50) During the period from timing t1 to timing t2, the channel interfaces 251_1 and 251_2 operate by using the power supplied from the regulator 252 via the power supply line 255, and the write commands WC1 and WC2 are transmitted to the nonvolatile memories 3_1 and 3_2 via the channels CH_1 and CH_2, respectively. At this time, the frequencies of the signals of the write commands WC1 and WC2 are smaller than, for example, a frequency of a signal of a write data WD. Operating frequencies of the channel interfaces 251_1 and 251_2 are also smaller than an operating frequency of a write data WD signal. Accordingly, the current supplied from the regulator 252 to the power supply line 255 is relatively small as illustrated in FIG. 2.

(51) Herein, since the current after timing t2 is relatively small, the current in the power supply line 255 is increased relatively gently between timings t1 and t2. In other words, the current change rate of the regulator 252 is relatively small and is to be the limit threshold value or less.

(52) At timing t3, when the transfer of the write commands WC1 and WC2 is ended, the channel interfaces 251_1 and 251_2 stop operating, and the current flowing through the power supply line 255 is decreased relatively gently. Accordingly, the current change rate of the regulator 252 is allowed to be relatively small and to be the limit threshold value or less.

(53) Next, at timing t4, the memory interface 25 almost simultaneously receives write data WD_1 for the nonvolatile memory 3_1 and the write data WD_2 for the nonvolatile memory 3_2. At this circuit time, the transfer 254 almost simultaneously transfers the write data WD_1 and WD_2 to the channel interfaces 251_1 and 251_2.

(54) During the period from timing t4 to timing t5, the channel interfaces 251_1 and 251_2 operate by using the power supplied from the regulator 252 via the power supply line 255, and the write data WD_1 and WD_2 are transmitted to the nonvolatile memory 3_1 and 3_2 via the channels CH_1 and CH_2, respectively. At this time, the frequency of the signals of the write data WD_1 and WD_2 is higher than the frequency of the signal of the write command WC. The operating frequencies of the channel interfaces 251_1 and 251_2 are also higher than the operating frequency of the write command WC. Accordingly, as indicated by the one-dot dashed line in FIG. 2, the current supplied from the regulator 252 to the power supply line 255 becomes relatively large.

(55) As indicated by the one-dot dashed line in FIG. 2, when the current supplied from the regulator 252 to the power supply line 255 is relatively large, since the current after the timing t5 is relatively large, the current of the power supply line 255 at timing t4 to timing t5 is increased relatively sharply. That is, the current change rate of the regulator 252 becomes relatively large and

exceeds the limit threshold value. When the current change rate of the regulator 252 exceeds the limit threshold value, the regulator 252 has difficulty in following the control operation so that the power supply voltage is equal to the target value. Accordingly, the current supplied to the power supply line 255 is allowed to be unstable. Accordingly, the channel interface 251 is likely to malfunction.

(56) Therefore, the control circuit 253 of this embodiment delays the start timing of the data transfer from the channel interfaces 251_1 and 251_2 to the nonvolatile memories 3_1 and 3_2 via the channels CH_1 and CH_2 by using the delay value parameter 41a. The control circuit 253 starts the data transfer via the channel CH_1 at timing t4, and the control circuit 253 starts the data transfer via the channel CH_2 at timing t6. The timing t6 is the timing delayed by the delay time ΔT_d converted from the delay value corresponding to the delay value parameter 41a.

(57) That is, during the period from timing t4 to timing t5, the channel interface 251_1 selectively operates by using the power supplied from the regulator 252 via the power supply line 255 to transfer the write data WD_1 to the nonvolatile memory 3_1 via the channel CH_1. At this time, since the operating channel interface is limited to the channel interface 251_1, as indicated by the solid line in FIG. 2, the current change rate supplied from the regulator 252 to the power supply line 255 can be reduced. In other words, the current change rate of the output current of the regulator 252 is allowed to be moderate.

(58) During the period between timing t5 and timing t6, the current supplied from the regulator 252 to the power supply line 255 is maintained substantially constant and stable.

(59) During the period from timing t6 to timing t7, the channel interface 251_2 additionally operates by using the power supplied from the regulator 252 via the power supply line 255, and the write data WD_2 is transferred to the nonvolatile memory 3_2 via the channel CH_2. At this time, the transfer of the write data WD_1 is continuously performed, and the channel interface 251_1 continuously performs the operations. By starting the operations of the channel interfaces 251 one by one, as indicated by the solid line in FIG. 2, the current change rate of the current supplied from the regulator 252 to the power supply line 255 can be reduced. In other words, the current change rate of the output current of the regulator 252 is allowed to be moderate.

(60) As indicated by the solid line in FIG. 2, when delaying the start timing of the data transfer from the channel interface 251 to the nonvolatile memory 3, for the current in the power supply line 255, since the current rise is reduced between timing t5 and timing t6, the current from the regulator 252 is increased relatively gently between timings t4 and t5. The current change rate of the regulator 252 can be reduced and, thus, can be smaller than the limit threshold value.

(61) Similarly, for the current in the power supply line 255, the difference between the current between timings t5 and t6 and the current after timing t7 becomes small compared to the case indicated by the one-dot dashed line. Along with this, the current from the regulator 252 at timings t6 to timing t7 is increased relatively gently. The current change rate in regulator 252 can be reduced and, thus, can be smaller than the limit threshold value. Accordingly, the regulator 252 can easily follow the control operation so that the power supply voltage becomes equal to the target value, and can allow the current in the power supply line 255 to be stabilized.

(62) Next, at timing t8, when the transfer of the write data WD_1 is completed, the channel interface 251_1 stops operating. At this time, the transfer of the write data WD_2 is continuously performed, and the channel interface 251_2 continuously performs the operations. The current flowing through the power supply line 255 is decreased relatively gently and is maintained constant at the predetermined current. Accordingly, the current change rate of the regulator 252 is relatively small and becomes the limit threshold value or less.

(63) At timing t9, when the transfer of the write data WD_2 is completed, the channel interface 251_2 stops operating, and the current flowing through the power supply line 255 is decreased relatively gently from the predetermined current. Accordingly, the current change rate of the regulator 252 is relatively small and becomes the limit threshold value or less.

(64) By delaying the start timing of the data transfer from the channel interface **251** to the nonvolatile memory **3**, the regulator **252** can easily follow the control operation so that the power supply voltage becomes equal to the target value. Accordingly, the current in the power supply line **255** during the write process of the memory system **1** can be stabilized.

(65) The controller **2** may determine the delay value to be converted into the delay time ΔT_d by performing the training. FIG. **3** is a flowchart illustrating the training operations of the memory system **1**. FIG. **3** illustrates the training of the operation when the memory interface **25** almost simultaneously receives the write data **WD_1** to **WD_N** for the nonvolatile memories **3_1** to **3_N**. The write data **WD_1** to **WD_N** may be dummy data for the training.

(66) It is noted that the training illustrated in FIG. **3** may be performed when the memory system **1** is started, may be performed when the memory system **1** is shipped, or may be performed at any timing when the memory system **1** is in operation.

(67) In the memory system **1**, the controller **2** searches for the delay value so that the current change rate of the regulator **252** is the limit threshold value or less.

(68) For example, the controller **2** starts the training illustrated in FIG. **3** when the predetermined condition is satisfied. The predetermined condition may be that the training execution request is received from the host **100** and the delay value parameter **41a** is not stored in the parameter buffer **41**. In this case, when the delay value parameter **41a** is already stored in the parameter buffer **41**, the training illustrated in FIG. **3** may be skipped.

(69) Alternatively, the predetermined condition may be receiving the training execution request from the host **100**. In this case, when the delay value parameter **41a** is already stored in the parameter buffer **41**, the controller **2** may erase the delay value parameter **41a** on the parameter buffer **41**.

(70) When starting the training, the controller **2** sets a delay value **D** to the initial value of “0” (**S1**).

(71) The controller **2** accesses the nonvolatile memories **3_1** to **3_N** by using the delay value **D=0** (**S2**). For example, for the write data **WD_1** to **WD_N**, the memory interface **25** delays the start timing of the data transfer from the channel interfaces **251_1** to **251_N** to the nonvolatile memories **3_1** to **3_N** via the channels **CH_1** to **CH_N** by the delay value **D=0**. That is, the memory interface **25** almost simultaneously transfers the write data **WD_1** to **WD_N** via the channels **CH_1** to **CH_N**.

(72) It is noted that the bit pattern of the data used for the training may be set to a toggle pattern in which 1 and 0 are repeated, for example, so that the current is increased.

(73) The controller **2** determines whether the **n** accesses to the nonvolatile memory **3** succeed for all the channels **CH_1** to **CH_N** that are training targets (**S3**). For example, for the write data **WD_1** to **WD_N**, when the controller **2** receives the write failure notification from the nonvolatile memory **3** via at least one channel **CH**, the controller **2** determines that there is a channel **CH** for which the access fails (No in **S3**).

(74) When there is a channel **CH** for which the access fails (No in **S3**), the controller **2** changes the delay value **D**. For example, the controller **2** sets the delay value **D** to **D+ ΔD** obtained by increasing the delay value **D** by the predetermined value ΔD (for example, 1) (**S4**).

(75) The controller **2** determines whether the delay value **D** is larger than or equal to the maximum allowable range value **Dlmt** (**S5**). The maximum allowable range value **Dlmt** is a value corresponding to the performance of the regulator **252**. The maximum allowable range value **Dlmt** may correspond to the limit threshold value for the current change rate of the regulator **252**. The maximum allowable range value **Dlmt** may correspond to the limit threshold value of the current change rate. The maximum allowable range value **Dlmt** may be a value which is the count value corresponding to the delay time during which the current change rate of the regulator **252** becomes smaller than the limit threshold value when the number of clocks is counted by the counter.

(76) The controller **2** compares the delay value **D** with the maximum allowable range value **Dlmt**. When the delay value **D** is larger than or equal to the maximum allowable range value **Dlmt** (Yes in

S5), the training is abnormally ended.

(77) Herein, when the training is performed before the shipment of the memory system **1**, the controller **2** determines that the regulator **252** is defective when the training ends abnormally. When the training is performed while the memory system **1** is in operation and the training is abnormally ended, the controller **2** determines deterioration of the regulator **252** and, after that, does not perform the operation of accessing the plurality of channels CH in parallel. Alternatively, the controller **2** notifies the host that the memory system **1** is in disorder.

(78) The controller **2** compares the delay value D with the maximum allowable range value D_{limt} . When the delay value D is smaller than the maximum allowable range value D_{limt} (No in S5), the process returns to S2.

(79) The loop of S2.fwdarw.S3.fwdarw.S4.fwdarw.S5.fwdarw.S2 is repeated until YES in S3 or YES in S5. For example, when the number of repetitions is n (n is an integer of 1 or more), the controller **2** sets the delay value $D=n \times \Delta D$ in the n -th S4. After that, when the delay value D is smaller than the maximum allowable range value D_{limt} (No in S5), the controller **2** accesses the nonvolatile memories **3_1** to **3_N** by using the delay value $D=n \times \Delta D$ (S2).

(80) For example, for the write data WD_1 to WD_N , the memory interface **25** delays the start timing of the data transfer from the channel interfaces **251_1** to **251_N** to the nonvolatile memories **3_1** to **3_N** via the channels CH_1 to CH_N by the delay value $D=n \times \Delta D$. That is, after the transfer of the write data WD_1 via the channel CH_1 is started, the memory interface **25** delays the transfer of the write data WD_2 via the channel CH_2 by the delay time ΔT_d converted from the delay value $D=n \times \Delta D$. After the transfer of the write data WD_2 via the channel CH_2 is started, the memory interface **25** delays the transfer of the write data WD_3 via the channel CH_3 by the delay time ΔT_d converted from the delay value $D=n \times \Delta D$. After the transfer of the write data $WD_{(N-1)}$ via the channel CH_{(N-1)} is started, the memory interface **25** delays the transfer of the write data WD_N via the channel CH_N by the delay time ΔT_d converted from the delay value $D=n \times \Delta D$.

(81) The controller **2** determines whether the access to the nonvolatile memory **3** succeeds for all the channels CH_1 to CH_N (S3). For example, for the write data WD_1 to WD_N , when the controller **2** receives the write completion notification from the nonvolatile memory **3** via all the channels CH_1 to CH_N, the controller **2** determines that the access to all the channels CH_1 to CH_N succeeds (Yes in S3).

(82) When the access to all the channels CH_1 to CH_N succeeds (Yes in S3), the controller **2** determines the current delay value D as the delay value that allows the current change rate of the regulator **252** to be the limit threshold value or less. The controller **2** generates the delay value parameter **41a** corresponding to the determined delay value D and writes the delay value parameter **41a** in the parameter buffer **41** of the buffer memory **4**. The controller **2** writes the delay value parameter **41a** in the management information storage area of the nonvolatile memory **3** so as to be nonvolatile (S6).

(83) Accordingly, when write access requests via the channels CH_1 and CH_2 are generated in parallel, the memory interface **25** reads and acquires the delay value parameter **41a** from the management information storage area of the nonvolatile memory **3**. The memory interface **25** can delay the start timing of the data transfer to the nonvolatile memories **3_1** and **3_2** via the channels CH_1 and CH_2 by the delay value corresponding to the delay value parameter **41a**. That is, the memory interface **25** can convert the delay value corresponding to the delay value parameter **41a** into the delay time and can delay the start timing of the data transfer to the nonvolatile memories **3_1** and **3_2** via the channels CH_1 and CH_2 by the converted delay time. Accordingly, the current change rate of the current supplied from the regulator **252** to the power supply line **255** can be reduced, and the current change rate can be allowed to be smaller than the limit threshold value.

(84) As described above, in the embodiment, when the two or more access requests to the nonvolatile memory **3** are generated in parallel, the controller **2** delays the timing of starting the

data transfer from the channel interface **251** to the nonvolatile memory **3** by the delay value corresponding to the delay value parameter **41a**. Accordingly, the control operation of the power supply current by the regulator **252** can be stabilized, and malfunction of the channel interface **251** receiving the power supply current can be prevented.

(85) It is noted that the delay value parameter **41a** may be stored in the eFUSE (not illustrated) instead of the nonvolatile memory **3**. The eFUSE is a memory that unrewritably stores initial parameters in the memory system **1**. When there is an unused area in the eFUSE, the delay value parameter **41a** can be allowed to be nonvolatile in the memory system **1** by storing the delay value parameter **41a** in the unused area. In this case, when two or more access requests to the nonvolatile memory **3** are generated in parallel, the controller **2** acquires the delay value parameter **41a** from the eFUSE.

(86) In addition, when the data transfer from the nonvolatile memory **3** to the memory interface **25** is performed, the read enable signal REZ may be used as the control signal for the channel CH for delaying the start timing. The read enable signal REZ is a clock used when processing the read request from the host. When the data is transferred from the memory interface **25** to the nonvolatile memory **3**, a data strobe signal DQS may be used as the control signal for the channel CH for delaying the start timing. The data strobe signal DQS is a clock used when processing the write request from the host. In the following embodiments and modified examples, steps including substantially the same processing as in FIG. **3** are denoted by the same step numbers. Different characters or numbers may be added to the end of the same step number when specifically distinguishing between steps including similar processing.

Modified Example 1 of First Embodiment

(87) In the first embodiment, the memory system **1** determines the delay value according to whether the access of the channel succeeds or fails. In Modified Example 1 of the embodiment, as illustrated in FIG. **4**, the memory system **1** may determine the delay value according to whether the level of the power supply voltage of the power supply line **255** is in an allowable range (whether the regulator **252** normally operates) instead of whether the access of the channel succeeds or fails. FIG. **4** is a flowchart illustrating training operations of the memory system **1** in Modified Example 1 of the first embodiment.

(88) It is noted that the training illustrated in FIG. **4** may be performed when the memory system **1** is started, may be performed when the memory system **1** is shipped, or may be performed at any timing when the memory system **1** is in operation.

(89) For example, the controller **2** starts training illustrated in FIG. **4** according to the fact that a predetermined condition is satisfied. The predetermined condition may be that the training execution request is received from the host **100** and the delay value parameter **41a** is not stored in the parameter buffer **41**. In this case, when the delay value parameter **41a** is already stored in the parameter buffer **41**, the training illustrated in FIG. **4** may be skipped.

(90) Alternatively, the predetermined condition may be receiving the training execution request from the host **100**. In this case, when the delay value parameter **41a** is already stored in the parameter buffer **41**, the controller **2** may erase the delay value parameter **41a** on the parameter buffer **41**.

(91) In the memory system **1**, the controller **2** performs operations differently from those of FIG. **3** in the following points. After performing the processes of S1 and S2, the controller **2** detects the level of the power supply line **255** and determines whether the level of the power supply voltage is in an allowable range (S3a). For example, when the allowable range is 2.7 V to 3.456 V, the controller **2** performs the process of S4 when the level of the power supply line **255** is attenuated outside the allowable range (2.7 V to 3.456 V) (No in S3a). When the level of the power supply line **255** falls in the allowable range (2.7 V to 3.456 V) (Yes in S3a), the controller **2** writes the delay value parameter **41a** to the management information storage area of the nonvolatile memory **3** so as to be nonvolatile (S6).

(92) Accordingly, when the write access requests via the channels CH_1 and CH_2 are generated in parallel, the memory interface 25 reads and acquires the delay value parameter 41a from the management information storage area of the nonvolatile memory 3.

(93) With such an operation, the delay value that allows the current change rate of the regulator 252 to be the limit threshold value or less can be determined. Accordingly, the control operation of the power supply current by the regulator 252 can be stabilized, and malfunction of the channel interface 251 receiving the power supply current can be prevented.

Modified Example 2 of First Embodiment

(94) Alternatively, although not illustrated, the controller 2 may be provided with a function (current measurement function) of measuring an operating current of the regulator 252. The memory interface 25 may incorporate the current measurement function.

(95) As Modified Example 2 of the first embodiment, as illustrated in FIG. 5, the memory system 1 may determine the delay value according to the current change rate instead of whether the access of the channel succeeds or fails. FIG. 5 is a flowchart illustrating training operations of the memory system 1 in Modified Example 2 of the first embodiment.

(96) It is noted that the training illustrated in FIG. 5 may be performed when the memory system 1 is started, may be performed when the memory system 1 is shipped, or may be performed at any timing when the memory system 1 is in operation.

(97) For example, the controller 2 starts training illustrated in FIG. 5 according to the fact that a predetermined condition is satisfied. The predetermined condition may be that the training execution request is received from the host 100 and the delay value parameter 41a is not stored in the parameter buffer 41. In this case, when the delay value parameter 41a is already stored in the parameter buffer 41, the training illustrated in FIG. 5 may be skipped.

(98) Alternatively, the predetermined condition may be receiving the training execution request from the host 100. In this case, when the delay value parameter 41a is already stored in the parameter buffer 41, the controller 2 may erase the delay value parameter 41a on the parameter buffer 41.

(99) In the memory system 1, the controller 2 performs operations differently from those of FIG. 3 in the following points. After performing the process of S2, the controller 2 detects the current flowing through the power supply line 255 and determines whether the current change rate is smaller than the limit threshold value (S3b). When the current change rate of the power supply line 255 is equal to or higher than the limit threshold value (No in S3b), the controller 2 performs the process of S4. When the current change rate of the power supply line 255 is smaller than the limit threshold value (Yes in S3b), the controller 2 writes the delay value parameter 41a in the management information storage area of the nonvolatile memory 3 so as to be nonvolatile (S6).

(100) Accordingly, when the write access requests via the channels CH_1 and CH_2 are generated in parallel, the memory interface 25 reads and acquires the delay value parameter 41a from the management information storage area of the nonvolatile memory 3.

(101) With such an operation, the delay value that allows the current change rate of the regulator 252 to be the limit threshold value or less can be determined. Accordingly, the control operation of the power supply current by the regulator 252 can be stabilized, and malfunction of the channel interface 251 receiving the power supply current can be prevented.

Modified Example 3 of First Embodiment

(102) In addition, as Modified Example 3 of the first embodiment, the operation of delaying the start timing of the data transfer by the delay value D may be implemented by the configuration and operations of the memory interface as illustrated in FIG. 6. FIG. 6 is a circuit diagram illustrating the configuration and operations of the memory interface 25 in Modified Example 3 of the first embodiment.

(103) As illustrated in FIG. 6, the memory interface 25 exchanges signals between the channel interfaces 251_1 to 251_N and the control circuit 253 when access requests via the plurality of

channels CH_1 to CH_N are generated in parallel, and thus, the start timing of the data transfer between the plurality of channels CH_1 to CH_N is adjusted.

(104) In Modified Example 3 of the first embodiment, the channel interface **251** transmits a start request SR requesting an instruction to start transferring the data to the nonvolatile memory **3** to the control circuit **253**. When receiving the start request SR from the channel interface **251**, the control circuit **253** acquires the delay value parameter **41a** from the nonvolatile memory **3**. At the predetermined timing, the control circuit **253** issues a start instruction SI corresponding to the start request SR to the channel interface **251**. The start instruction SI is a signal that instructs the start of the data transfer.

(105) Specifically, the control circuit **253** includes a first-in-first-out (FIFO) circuit **2531**, an issue circuit **2532**, and a counter **2533**.

(106) The issue circuit **2532** stores the start request SR transmitted by the channel interface **251** in the FIFO circuit **2531**. The FIFO circuit **2531** has a plurality of stages of queues. When receiving the start request SR, the issue circuit **2532** sequentially stores the start request SR from the head queue of the FIFO circuit **2531**.

(107) The issue circuit **2532** adjusts the start timing of the data transfer between the plurality of channel interfaces **251_1** to **251_N** according to one or more start requests SR stored in the FIFO circuit **2531**. The issue circuit **2532** fetches the start request SR from the head queue of the FIFO circuit **2531**. The FIFO circuit **2531** shifts the stored start request SR to the head side queue every time the start request SR is fetched from the head queue. The issue circuit **2532** issues the start instruction SI according to the fetched start request SR.

(108) The counter **2533** receives a predetermined reference clock CLK and counts the number of clocks. When the issue circuit **2532** issues the start instruction SI, the counter **2533** starts counting operation. When the count value reaches the delay value D, the counter **2533** notifies the issue circuit **2532** of the fact that the count value reaches the delay value D (count completion notification).

(109) When the issue circuit **2532** receives the count completion notification, the issue circuit **2532** fetches the start request SR from the head queue of the FIFO circuit **2531** again. The issue circuit **2532** issues the start instruction SI according to the fetched start request SR.

(110) For example, the write command and the write data are transferred from the transfer circuit **254** to the channel interfaces **251_1** and **251_2**. The channel interface **251_1** transmits a start request SR1 to the control circuit **253**. The channel interface **251_2** transmits a start request SR2 to the control circuit **253**. When receiving the start requests SR1 and SR2, the control circuit **253** acquires the delay value parameter **41a** from the nonvolatile memory **3** and stores the delay value parameter **41a** in the parameter buffer **41**. At first timing, the control circuit **253** issues a start instruction SI1 to the channel interface **251_1** according to the start request SR1. At second timing delayed by the delay value D from the first timing, the control circuit **253** issues a start instruction SI2 according to the start request SR2 to the channel interface **251_2**.

(111) Specifically, the channel interfaces **251_1** and **252_2** almost simultaneously receive write data WD1 and WD2 from transfer circuit **254**. That is, the write access requests via the channels CH_1 and CH_2 are generated in parallel. When receiving the write data WD1, the channel interface **251_1** transmits a start signal CH_1_Start to the control circuit **253** as the start request SR. In parallel, according to receiving the write data WD2, the channel interface **251_2** transmits a start signal CH_2_Start to the control circuit **253** as the start request SR.

(112) The control circuit **253** determines that the start signal CH_1_Start is to be first processed among the start signal CH_1_Start and the start signal CH_2_Start based on the predetermined specification. The predetermined specification may be to process the start signals on a first-come-first-served basis.

(113) The control circuit **253** transmits a non-active level mask signal CH_1_Mask to the channel interface **251_1**. At the same time, the control circuit **253** transmits the active level mask signal

CH_1_Mask to the channel interface 251_2. The channel interface 251 waits for the data transfer when the mask signal is at the active level. The channel interface 251 starts the data transfer when the mask signal transitions from the active level to the non-active level.

(114) At timing t4, the channel interface 251_1 starts transferring the write data WD_1 to the nonvolatile memory 3_1 via the channel CH_1 according to the non-active level mask signal CH_1_Mask.

(115) On the other hand, the channel interface 251_2 waits without transferring the write data WD_2 according to an active level mask signal CH_2_Mask.

(116) Next, the control circuit 253 measures the delay time ΔT_d converted from the delay value D. At timing t6 after the delay time ΔT_d has elapsed from timing t4, the control circuit 253 allows the mask signal CH_2_Mask to transition from the active level to the non-active level. That is, the control circuit 253 issues the non-active level mask signal CH_2_Mask to the channel interface 251_2 as the start instruction for instructing the start of the data transfer. According to the non-active level mask signal CH_2_Mask, the channel interface 251_2 starts transferring the write data WD_2 to the nonvolatile memory 3_2 via the channel CH_2.

(117) That is, during the period from timing t4 to timing t5, the channel interface 251_1 selectively operates by using the power supplied from the regulator 252 via the power supply line 255 to transfer the write data WD_1 to the nonvolatile memory 3_1 via the channel CH_1. At this time, since the operating channel interface is limited to the channel interface 251_1, as indicated by a solid line in FIG. 2, the change rate of the current supplied from the regulator 252 to the power supply line 255 can be reduced.

(118) During the period from timing t6 to timing t7, the channel interface 251_2 additionally operates by using the power supplied from the regulator 252 via the power supply line 255 and transfers the write data WD_2 to the nonvolatile memory 3_2 via the channel CH_2. At this time, since the channel interface that additionally operates is limited to the channel interface 251_2, as indicated by a solid line in FIG. 2, the change rate of the current supplied from the regulator 252 to the power supply line 255 can be reduced.

(119) FIG. 7 is a waveform diagram illustrating an example of the operations of the memory interface 25 in Modified Example 3 of the first embodiment. For example, at timing t11, the control circuit 253 receives the start signals CH_1_Start and CH_2_Start in order from the channel interfaces 251_1 and 251_2. The control circuit 253 reads the delay value parameter 41a from the nonvolatile memory 3.

(120) At timing t12, the FIFO circuit 2531 puts the start signal CH_1_Start received first among the start signals CH_1_Start and CH_2_Start into the head queue of the FIFO circuit 2531. The FIFO circuit 2531 puts the next received start signal CH_1_Start into the queue of the next stage. The issue circuit 2532 collectively transitions the mask signals CH_1_Mask to CH_N_Mask of each channel to the active level. The issue circuit 2532 sets the delay value D (for example, D=8) according to the delay value parameter 41a in the counter 2533 as the count value for ending counting.

(121) At timing t13, the issue circuit 2532 fetches the start signal CH_1_Start from the head queue of the FIFO circuit 2531. The FIFO circuit 2531 shifts the start signal CH_2_Start from the next-stage queue to the head queue. The issue circuit 2532 allows the mask signal CH_1_Mask to transition from the active level to the non-active level according to the extracted start signal CH_1_Start. The counter 2533 starts counting the number of clocks of the reference clock CLK.

(122) At timing t4, the channel interface 251_1 starts toggling the data strobe signal CH_1 DQS according to the non-active level of the mask signal CH_1_Mask, and starts the transfer of the write data WD_1 to the nonvolatile memory 3_1 via the channel CH_1.

(123) At timing t14, when the count value reaches the delay value D, the counter 2533 transmits the count completion notification to the FIFO circuit 2531. After that, the counter 2533 resets the count value.

(124) At timing **t15**, the issue circuit **2532** fetches the start signal **CH_2_Start** from the head queue of the FIFO circuit **2531**. The issue circuit **2532** allows the mask signal **CH_2_Mask** to transition from the active level to the non-active level according to the extracted start signal **CH_2_Start**. The counter **2533** restarts counting the number of clocks of the reference clock **CLK**.

(125) At timing **t6**, the channel interface **251_2** starts toggling the data strobe signal **CH_2_DQS** according to the non-active level of the mask signal **CH_2_Mask**, and starts the transfer of the write data **WD_2** to the nonvolatile memory **3_2** via the channel **CH_2**.

(126) With such the configuration and operations, when the access requests via the plurality of channels **CH_1** to **CH_N** are generated in parallel, the memory interface **25** can delay the start timing of the data transfer between the plurality of channels **CH_1** to **CH_N** by the delay value **D**.

(127) Accordingly, the current change rate of the current supplied from the regulator **252** to the power supply line **255** can be reduced, and the current change rate can be allowed to be smaller than the limit threshold value. Accordingly, the control operation of the power supply current by the regulator **252** can be stabilized, and malfunction of the channel interface **251** receiving the power supply current can be reduced.

Second Embodiment

(128) In the first embodiment, when the plurality of channels are accessed in parallel, the delay value is determined so that the current change rate of the regulator **252** becomes the limit threshold value or less. Compared to the first embodiment, the second embodiment determines the number of channels to which the access can be performed in parallel instead of determining the delay time.

(129) Since the configuration of the memory system of the second embodiment is the same as that of the memory system of the first embodiment, the same configurations of the memory system of the second embodiment as those of the memory system of the first embodiment are denoted by the same reference numerals, and the description thereof is omitted.

(130) When the controller **2** accesses the plurality of channels in parallel, the training is performed to find the number of channels for which the current change rate of the regulator **252** is smaller than the limit threshold value.

(131) FIG. **8** is a flowchart illustrating the training operation of the memory system **1** according to the second embodiment. FIG. **8** illustrates operation training when the memory interface **25** almost simultaneously receives the write data **WD_1** to **WD_N** for the nonvolatile memories **3_1** to **3_N**.

(132) It is noted that the training illustrated in FIG. **8** may be performed when the memory system **1** is started, may be performed when the memory system **1** is shipped, or may be performed at any timing when the memory system **1** is in operation.

(133) For example, the controller **2** starts the training illustrated in FIG. **8** when the predetermined condition is satisfied. The predetermined condition may be that the training execution request is received from the host **100** and the parallel channel number parameter **41b** is not stored in the parameter buffer **41**. In this case, when the parallel channel number parameter **41b** is already stored in the parameter buffer **41**, the training illustrated in FIG. **8** may be skipped.

(134) Alternatively, the predetermined condition may be receiving the training execution request from the host **100**. In this case, when the parallel channel number parameter **41b** is already stored in the parameter buffer **41**, the controller **2** may erase the parallel channel number parameter **41b** on the parameter buffer **41**.

(135) When the controller **2** starts training for the number of parallel channels, the controller **2** sets the initial value **M0** to the number **M** of parallel channels (**S11**). **M0=8** may be used.

(136) The controller **2** uses the number **M** of parallel channels as **M=M0** to access **M** nonvolatile memories **3** among the **N** nonvolatile memories **3_1** to **3_N** in parallel (**S12**). For example, for the write data **WD_1** to **WD_M**, the memory interface **25** starts the data transfer from the channel interfaces **251_1** to **251_M** to the nonvolatile memories **3_1** to **3_M** via the channels **CH_1** to **CH_M** at the same timing. Accordingly, the memory interface **25** performs the transfer of the write data **WD_1** to **WD_M** via the channels **CH_1** to **CH_M** in parallel.

(137) The controller 2 determines whether the access to the nonvolatile memory 3 succeeds for all the channels CH_1 to CH_M that are a training target (S13). For example, for the write data WD_1 to WD_M, when the controller 2 receives the write failure notification from the nonvolatile memory 3 via at least one channel CH, the controller 2 determines that there is a channel CH for which the access fails (No in S13).

(138) When there is a channel CH for which the access fails (No in S13), the controller 2 changes the number M of parallel channels. For example, the controller 2 sets $M - \Delta M$ obtained by subtracting the predetermined number ΔM (for example, 1) from the number M of parallel channels to the number M of parallel channels (S14).

(139) The controller 2 determines whether the number M of parallel channels is larger than a minimum allowable channel number Mth (S15). The minimum allowable channel number Mth=1 may be used.

(140) The controller 2 compares the number M of parallel channels with the minimum allowable channel number Mth. When the number M of parallel channels is the minimum allowable channel number Mth or less (No in S15), the controller 2 abnormally ends the training.

(141) The controller 2 compares the number M of parallel channels with the minimum allowable channel number Mth. When the number M of parallel channels is larger than the minimum allowable channel number Mth (Yes in S15), the controller 2 returns the process to S12.

(142) The controller 2 determines whether the access to the nonvolatile memory 3 succeeds for all the channels CH_1 to CH_M that are training targets (S13). For example, for the write data WD_1 to WD_M, when the controller 2 receives the write completion notification from the nonvolatile memory 3 via all the channels CH_1 to CH_M, the controller 2 determines that the access to all the channels CH_1 to CH_M succeeds (Yes in S13).

(143) When the access to all the channels CH_1 to CH_M succeeds (Yes in S13), the controller 2 determines the current number M of parallel channels as the parallel channel number that allows the current change rate of the regulator 252 to be the limit threshold value or less. The controller 2 generates the parallel channel number parameter 41b corresponding to the determined number M of parallel channels on the buffer memory 4 and stores the parallel channel number parameter 41b in the parameter buffer 41. The controller 2 writes the parallel channel number parameter 41b in the management information storage area of the nonvolatile memory 3 so as to be nonvolatile (S16).

(144) Accordingly, when the write access requests via the channels CH_1 to CH_N are generated in parallel, the memory interface 25 reads and acquires the parallel channel number parameter 41b from the management information storage area of the nonvolatile memory 3. The memory interface 25 can limit the number of channels for performing the data transfer in parallel to the number M of channels according to the parallel channel number parameter 41b. Accordingly, the current change rate of the current supplied from the regulator 252 to the power supply line 255 can be reduced to be the limit threshold value or less, and deterioration of the regulator 252 can be prevented. In addition, the embodiment can cope with aging degradation of the regulator 252 due to long-term operation of the system, individual differences in chip characteristics, and the like. In addition, the embodiment can flexibly cope with the change in the number of channels mounted on the memory interface 25.

Modified Example 1 of Second Embodiment

(145) As Modified Example 1 of the second embodiment, as illustrated in FIG. 9, the memory system 1 may perform training on a data bus width of each number of channels. FIG. 9 is a flowchart illustrating training operations of the memory system 1 according to Modified Example 1 of the second embodiment. FIG. 9 illustrates an example of the training of the operation when the memory interface 25 almost simultaneously receives the write data WD_1 to WD_N for the nonvolatile memories 3_1 to 3_N.

(146) It is noted that the training illustrated in FIG. 9 may be performed when the memory system 1 is started, may be performed when the memory system 1 is shipped, or may be performed at any

timing when the memory system **1** is in operation.

(147) For example, the controller **2** starts the training illustrated in FIG. **9** when the predetermined condition is satisfied. The predetermined condition may be that the training execution request is received from the host **100** and the data bus width parameter **41c** is not stored in the parameter buffer **41**. In this case, when the data bus width parameter **41c** is already stored in the parameter buffer **41**, the training illustrated in FIG. **9** may be skipped.

(148) Alternatively, the predetermined condition may be receiving the training execution request from the host **100**. In this case, when the data bus width parameter **41c** is already stored in the parameter buffer **41**, the controller **2** may erase the data bus width parameter **41c** on the parameter buffer **41**.

(149) In the memory system **1**, the controller **2** swings the data bus width for each number of channels and searches for the data bus width that allows the current change rate of the regulator **252** to be the limit threshold value or less.

(150) When starting the training of the data bus width, the controller **2** sets the data bus width W to the initial value W_0 (S21). W_0 may be 16 (bits).

(151) The controller **2** accesses the N nonvolatile memories **3_1** to **3_N** by using the data bus width $W=W_0$ (S22). For example, for the write data WD_1 to WD_N , the memory interface **25** delays, by the delay value D , the timing at which the data transfer from the channel interfaces **251_1** to **251_N** to the nonvolatile memories **3_1** to **3_N** via the channels CH_1 to CH_N is started. Accordingly, the memory interface **25** transfers the write data WD_1 to WD_N via the channels CH_1 to CH_N in parallel while changing the timing of starting the transfer of the write data to each channel.

(152) The controller **2** determines whether the access to the nonvolatile memory **3** succeeds for all the channels CH_1 to CH_N that are training targets (S23). For example, for the write data WD_1 to WD_N , when the controller **2** receives the write failure notification from the nonvolatile memory **3** via at least one channel CH , the controller **2** determines that there is a channel CH for which the access fails (No in S23).

(153) When there is a channel CH for which the access fails (No in S23), the controller **2** changes the data bus width W . For example, the controller **2** sets to the data bus width W $W-\Delta W$ obtained by subtracting the predetermined number ΔW (for example, 4 (bits)) from the data bus width W (S24).

(154) The controller **2** determines whether the data bus width W is larger than the minimum allowable width W_{th} (S25). The minimum allowable width W_{th} may be 4 (bits).

(155) The controller **2** compares the data bus width W with the minimum allowable width W_{th} . When the data bus width W is smaller than the minimum allowable width W_{th} (No in S25), the controller **2** abnormally ends the training.

(156) The controller **2** compares the data bus width W with the minimum allowable width W_{th} . When the data bus width W is the minimum allowable width W_{th} or more (Yes in S25), the controller **2** returns the process to S22.

(157) The controller **2** determines whether the access to the nonvolatile memory **3** succeeds for all the channels CH_1 to CH_N that are training targets (S23). For example, for the write data WD_1 to WD_N , when the controller **2** receives the write completion notification from the nonvolatile memory **3** via all the channels CH_1 to CH_N , the controller **2** determines that the access to all the channels CH_1 to CH_N succeeds (Yes in S23).

(158) When the access to all the channels CH_1 to CH_N succeeds (Yes in S23), the controller **2** determines the current data bus width W as the data bus width that allows the current change rate of the regulator **252** to be the limit threshold value or less. The controller **2** generates the data bus width parameter **41c** corresponding to the determined data bus width W on the buffer memory **4** and stores the data bus width parameter **41c** in the parameter buffer **41**. The controller **2** stores the data bus width parameter **41c** in the management information storage area of the nonvolatile

memory **3** (S26).

(159) Accordingly, when the write access requests via the channels CH_1 to CH_N are generated in parallel, the memory interface **25** reads and acquires the data bus width parameter **41c** from the management information storage area of the nonvolatile memory **3**. The memory interface **25** can limit the data bus width for performing the data transfer in parallel to the data bus width W according to the data bus width parameter **41c**. Accordingly, the current change rate of the current supplied from the regulator **252** to the power supply line **255** can be reduced to be the limit threshold value or less, and deterioration of the regulator **252** can be reduced. In addition, it is possible to cope with aging degradation of the regulator **252** due to long-term operation of the system, individual differences in chip characteristics, and the like. In addition, it is possible to flexibly cope with the change in the number of channels mounted on the memory interface **25**.

Third Embodiment

(160) In addition, as a third embodiment, as illustrated in FIG. **10**, the memory system **1** may perform the training on the frequency of each channel CH. FIG. **10** is a flowchart illustrating the training operations of the memory system **1** according to the third embodiment. FIG. **10** illustrates the training of the operation when the memory interface **25** almost simultaneously receives the write data WD_1 to WD_N for the nonvolatile memories **3_1** to **3_N**.

(161) It is noted that the training illustrated in FIG. **10** may be performed when the memory system **1** is started, may be performed when the memory system **1** is shipped, or may be performed at any timing when the memory system **1** is in operation.

(162) For example, the controller **2** starts the training illustrated in FIG. **10** when the predetermined condition is satisfied. The predetermined condition may be that the training execution request is received from the host **100** and the frequency parameter **41d** is not stored in the parameter buffer **41**. In this case, when the frequency parameter **41d** is already stored in the parameter buffer **41**, the training illustrated in FIG. **10** may be skipped.

(163) Alternatively, the predetermined condition may be receiving the training execution request from the host **100**. In this case, when the frequency parameter **41d** is already stored in the parameter buffer **41**, the controller **2** may erase the frequency parameter **41d** on the parameter buffer **41**.

(164) Since the configuration of the memory system of the second embodiment is the same as that of the memory system of the first embodiment, the same configurations of the memory system of the second embodiment as those of the memory system of the first embodiment are denoted by the same reference numerals, and the description thereof is omitted.

(165) In the memory system **1**, the controller **2** swings the frequency of each channel CH and searches for the frequency that allows the current change rate of the regulator **252** to be the limit threshold value or less. The frequency of each channel CH is a frequency of the clock (for example, read enable signal REZ, data strobe signal DQS, or the like) used for the data signal of each channel CH.

(166) When the training of the frequency is started, the controller **2** sets to the frequency F the initial value F0 (S31). F0 may be 300 (MHz).

(167) The controller **2** accesses the nonvolatile memories **3** of the N nonvolatile memories **3_1** to **3_N** by using the frequency F=F0 (S32). For example, for the write data WD_1 to WD_N, the memory interface **25** delays, by the delay value D, the timing of starting the data transfer from the channel interfaces **251_1** to **251_N** to the nonvolatile memories **3_1** to **3_N** via the channels CH_1 to CH_N. Accordingly, the memory interface **25** transfers the write data WD_1 to WD_N via the channels CH_1 to CH_N in parallel.

(168) The controller **2** determines whether the access to the nonvolatile memory **3** succeeds for all the channels CH_1 to CH_N that are training targets (S33). For example, for the write data WD_1 to WD_N, when the controller **2** receives the write failure notification from the nonvolatile memory **3** via at least one channel CH, the controller **2** determines that there is a channel CH for

which the access fails (No in S33).

(169) When there is a channel CH for which the access fails (No in S33), the controller 2 changes the frequency F. For example, the controller 2 sets to the frequency $F - \Delta F$ obtained by subtracting the predetermined value ΔF (for example, 50 (MHz)) from the frequency F (S34).

(170) The controller 2 determines whether the frequency F is larger than the minimum allowable frequency F_{th} (S35). The minimum allowable frequency F_{th} may be 150 (MHz).

(171) The controller 2 compares the frequency F with the minimum allowable frequency F_{th} . When the frequency F is smaller than the minimum allowable frequency F_{th} (No in S35), the controller 2 abnormally ends the training.

(172) The controller 2 compares the frequency F with the minimum allowable frequency F_{th} . When the frequency F is the minimum allowable frequency F_{th} or more (Yes in S35), the controller 2 returns the process to S32.

(173) The controller 2 determines whether the access to the nonvolatile memory 3 succeeds for all the channels CH_1 to CH_N that are training targets (S33). For example, for the write data WD_1 to WD_N, when the controller 2 receives the write completion notification from the nonvolatile memory 3 via all the channels CH_1 to CH_N, the controller 2 determines that the access to all the channels CH_1 to CH_N succeeds (Yes in S33).

(174) When the access to all the channels CH_1 to CH_N succeeds (Yes in S33), the controller 2 determines the current frequency F as the frequency that allows the current change rate of the regulator 252 to be the limit threshold value or less. The controller 2 generates the frequency parameter 41d corresponding to the determined frequency F on the buffer memory 4 and stores the frequency parameter 41d in the parameter buffer 41. The controller 2 writes the frequency parameter 41d in the management information storage area of the nonvolatile memory 3 so as to be nonvolatile (S26).

(175) Accordingly, the memory interface 25 reads and acquires the frequency parameter 41d from the management information storage area of the nonvolatile memory 3 when the write access requests via the channels CH_1 to CH_N are generated in parallel. The memory interface 25 can limit the frequency of the clocks (for example, read enable signal REZ, data strobe signal DQS, or the like) during the data transfer to the frequency F or less according to the frequency parameter 41d. Accordingly, the current change rate of the current supplied from the regulator 252 to the power supply line 255 can be reduced to be the limit threshold value or less, and deterioration of the regulator 252 can be reduced. In addition, it is possible to cope with aging degradation of the regulator 252 due to long-term operation of the system, individual differences in chip characteristics, and the like. In addition, it is possible to flexibly cope with the change in the number of channels mounted on the memory interface 25.

(176) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A memory system comprising: a plurality of nonvolatile memories; and a controller connected to the plurality of nonvolatile memories via a plurality of channels, wherein a first nonvolatile memory of the plurality of nonvolatile memories stores a first parameter, wherein the controller is configured to acquire the first parameter from the first nonvolatile memory and delay start timing of data transfer to the first nonvolatile memory via a channel among the plurality of channels by a

delay value that is based on the first parameter, wherein the controller includes: a plurality of channel interfaces corresponding to the plurality of channels, each channel interface connected to the corresponding channel; a regulator configured to supply a current to the plurality of channel interfaces; and a control circuit configured to delay the start timing of the data transfer from the channel interface to a corresponding one of the nonvolatile memories via the channel by the delay value, and wherein the first parameter is based on performance of the regulator, wherein the plurality of channel interfaces include a first channel interface and a second channel interface, and wherein, when receiving a first start request from the first channel interface and receiving a second start request from the second channel interface, the control circuit issues a first start instruction according to the first start request at a first timing and issues a second start instruction according to the second start request at a second timing delayed from the first timing by the delay value.

2. The memory system according to claim 1, wherein, when access requests involving the data transfer to two or more nonvolatile memories of the plurality of nonvolatile memories via two or more channels among the plurality of channels are generated in parallel, the controller is configured to acquire the first parameter from the nonvolatile memory and delay the start timing of the data transfer to the nonvolatile memory via the channel among the two or more channels by the delay value.

3. The memory system according to claim 1, wherein the delay value according to the first parameter is determined so that a current change rate from the regulator is smaller than a threshold value.

4. The memory system according to claim 1, wherein the delay value according to the first parameter is determined according to a threshold value of a current change rate capable of being followed by a control operation of the regulator.

5. The memory system according to claim 1, wherein each of the plurality of channel interfaces has a buffer memory.

6. The memory system according to claim 1, wherein the controller is configured to access the nonvolatile memory and perform training to determine the first parameter when the memory system is shipped.

7. The memory system according to claim 1, wherein the controller is configured to access the nonvolatile memory and perform training to determine the first parameter when the memory system is started.

8. The memory system according to claim 1, wherein the controller is configured to access the nonvolatile memory and perform training to determine the first parameter at any timing when the memory system is in operation.

9. The memory system according to claim 1, wherein the controller is configured to specify the number of parallel channels successfully accessing the nonvolatile memory while changing the number of parallel channels capable of operating in parallel when performing the data transfer from the channel interface to the nonvolatile memory via the channel, generate a second parameter based on the specified number of parallel channels and store the second parameter in the nonvolatile memory, and wherein the control circuit is configured to perform the data transfer from the channel interface to the nonvolatile memory via the channel with the number of parallel channels according to the second parameter.

10. The memory system according to claim 9, wherein the controller is configured to access the nonvolatile memory and perform training to determine the second parameter at any timing of when the memory system is shipped, started, or in operation.

11. The memory system according to claim 1, wherein the controller is configured to specify a data bus width successfully accessing the nonvolatile memory while changing the data bus width used when performing the data transfer from the channel interface to the nonvolatile memory via the channel, generate a third parameter based on the specified data bus width, and store the third parameter in the nonvolatile memory, and wherein the control circuit is configured to perform the

data transfer from the channel interface to the nonvolatile memory via the channel with the data bus width according to the third parameter.

12. The memory system according to claim 11, wherein the controller is configured to access the nonvolatile memory and perform training to determine the third parameter at any timing of when the memory system is shipped, started, or in operation.

13. The memory system according to claim 1, wherein the controller is configured to specify a frequency successfully accessing the nonvolatile memory while changing the frequency when performing the data transfer from the channel interface to the nonvolatile memory via the channel, generate a fourth parameter based on the specified frequency, and store the fourth parameter in the nonvolatile memory, and wherein the control circuit is configured to operate at a frequency according to the fourth parameter and to perform the data transfer from the channel interface to the nonvolatile memory via the channel.

14. The memory system according to claim 13, wherein the controller is configured to access the nonvolatile memory and perform training to determine the fourth parameter at any timing of when the memory system is shipped, started, or in operation.

15. A memory system, comprising: a plurality of nonvolatile memories; and a controller connected to the plurality of nonvolatile memories via a plurality of channels, wherein a first nonvolatile memory of the plurality of nonvolatile memories stores a first parameter, wherein the controller is configured to acquire the first parameter from the first nonvolatile memory and delay start timing of data transfer to the first nonvolatile memory via a channel among the plurality of channels by a delay value that is based on the first parameter, wherein the controller includes: a plurality of channel interfaces corresponding to the plurality of channels, each channel interface connected to the corresponding channel; a regulator configured to supply a current to the plurality of channel interfaces; and a control circuit configured to delay the start timing of the data transfer from the channel interface to a corresponding one of the nonvolatile memories via the channel by the delay value, and wherein the first parameter is based on performance of the regulator, and wherein the control circuit includes: a first-in-first-out (FIFO) circuit configured to be input with a start request from the channel interface; an issue circuit configured to issue a first start instruction to the channel interface based on the first start request fetched from the FIFO circuit; and a counter configured to start a counting operation according to issuance of a start instruction by the issue circuit and to count up to the delay value, and wherein the issue circuit is configured to issue a second start instruction according to a second start request next fetched from a top of the FIFO circuit when the counter counts up to the delay value, to the channel interface.

16. A memory system, comprising: a plurality of nonvolatile memories; and a controller connected to the plurality of nonvolatile memories via a plurality of channels, wherein a first nonvolatile memory of the plurality of nonvolatile memories stores a first parameter, wherein the controller is configured to acquire the first parameter from the first nonvolatile memory and delay start timing of data transfer to the first nonvolatile memory via a channel among the plurality of channels by a delay value that is based on the first parameter, wherein the controller includes: a plurality of channel interfaces corresponding to the plurality of channels, each channel interface connected to the corresponding channel; a regulator configured to supply a current to the plurality of channel interfaces; and a control circuit configured to delay the start timing of the data transfer from the channel interface to a corresponding one of the nonvolatile memories via the channel by the delay value, and wherein the first parameter is based on performance of the regulator, and wherein the controller is configured to specify the delay value successfully accessing the nonvolatile memory while changing the delay value shifting the start timing of the data transfer from the channel interface to the nonvolatile memory via the channel, generate the first parameter based on the specified delay value, and store the first parameter in the nonvolatile memory.
